

Ran He

Ph.D. Candidate

The Pennsylvania State University | Department of Geosciences

437 Deike Building, University Park, PA 16802

(814) 996-8550 | heran@psu.edu | sites.psu.edu/carbonatehr

RESEARCH INTERESTS

Paleoclimate & paleoenvironment; Sedimentary basin evolution; Carbonate sedimentology & diagenesis; Low-temperature stable isotope geochemistry; Carbonate clumped isotope geochemistry

EDUCATION

The Pennsylvania State University

2021–present (expected 2026.12)

Ph.D. Candidate in Geosciences (GPA: 3.98/4.00)

Thesis: Siderite oxygen and clumped isotopes as an archive of terrestrial paleoclimate and diagenesis

Advisor: Dr. Miquela Ingalls

Committee members: Dr. Katherine Freeman, Dr. Brian Currie, Dr. Max Lloyd, Dr. William Burgos

China University of Geosciences (Beijing)

2016–2020

B.S. *Cum Laude* in Geology (GPA: 3.72/4.00, Rank 1/86)

Thesis: Marine $\delta^{13}\text{C}_{\text{DIC}}$ rise and primary productivity recovery after the Marinoan Snowball Earth

Advisor: Dr. Bing Shen (Peking University)

PROFESSIONAL APPOINTMENTS

Chengdu University of Technology

2020–2021

Research Assistant

Mentor: Dr. Xianguo Lang

RESEARCH PROJECTS

Carbonate Diagenesis

#3 Siderite clumped isotope (Δ_{47}) solid-state reordering kinetics

Ongoing

Using Pennsylvanian-age siderite-bearing rocks in the Appalachian Basin (PA, WV, KY, OH) to investigate how siderite and calcite clumped isotopes (Δ_{47}) behave in a series of high burial temperatures (near-greenschist phase metamorphism) and duration, compiling existing and new thermal history data, and incorporate them into siderite (Δ_{47}) alteration.

#2 Siderite stable isotopes systematics during low-T fluid-mediated diagenesis

Ongoing

Using petrographic and geochemical comparison on core-outcrop terrestrial siderite samples of the Paleocene Fort Union Formation, Wind River Basin, WY, to investigate the carbon, oxygen, and clumped isotopic behavior during fluid-mediated burial and modern chemical weathering.

#1 Mg isotope systematics of early diagenetic aragonite-calcite transition

Using the Pleistocene Key Largo coral samples, which include primary aragonite and diagenetic calcite, to investigate the Mg isotope behavior during meteoric and marine diagenesis. Results indicate the diagenetic calcite records a new Mg isotope signal of calcite neomorphism.

Marine Biogeochemistry

#2 Marine sulfate concentration reconstruction at Permian-Triassic boundary

Ongoing

Analyzing pyrite $\delta^{34}\text{S}_{\text{py}}$ of P-T boundary at Shangsi Section (deep marine shale), and using mathematic modeling to reconstruct marine sulfate concentration at the P-T boundary.

#1 Marine carbon cycle perturbation after the collapse of end-Cryogenian Marinoan Snowball Earth

Compiling carbon isotope data from the Marinoan Cap Carbonates and using mathematic modeling to unravel the perturbation of the carbon cycle during and after the Marinoan Glaciation. Results indicate that marine DIC after the Marinoan Glaciation are more enriched in ^{13}C than glaciation because of the recovery of primary productivity and organic carbon burial.

PUBLICATIONS

9. He, R., Lloyd, M., Currie, B., Ingalls, M. Iron carbonate stable isotopes are more robust to fluid-mediated alteration than calcium carbonate. (Manuscript in preparation for *Geology*)
8. Kalmer, M., Currie, B., He, R. Relationships between Vitrinite Reflectance and Programmed Pyrolysis Tmax Temperatures of Pennsylvanian Rocks in the Central and Northern Appalachian Basin. (Manuscript under review at *International Journal of Coal Geology*)
7. He, R., Lloyd, M., Currie, B., Ingalls, M. Facies-specific siderite clumped isotopes indicate subtropical climate in the Paleocene mid-latitude continental interior. (In press, *Journal of Sedimentary Research*)
6. Hartke, E., et al., including He, R. (2026) Confirmation of Δ^2 -Sterenes as Precursors of 3-Methylsteranes. *Organic Geochemistry*, 105231.
5. Wyman, A., et al., including He, R. (2025) Kiritimati Lagoon temperature reconstruction from clumped isotopes. *The Depositional Record*, 11(5), pp. 1479–1496.
4. Shen, B., et al., including He, R. (2025) Resolving the Snowball Earth conundrum: the role of marine dissolved organic carbon pool. *Science Bulletin*, 70(13), 2173–2189. (review paper)
3. Zhao, K., et al., including He, R. (2022) Low marine sulfate levels during the initiation of the Cryogenian Marinoan glaciation. *Precambrian Research*, 377, 106737.
2. He, R., Lang, X.G., and Shen, B. (2021) A rapid rise of seawater $\delta^{13}\text{C}$ during the deglaciation of the Marinoan Snowball Earth. *Global and Planetary Change*, 207, 103672.
1. ¹He, R., ¹Ning, M., Huang, K.J., Ma, H.R., and Shen, B. (2020) Mg isotopic systematics during early diagenetic aragonite-calcite transition: Insights from the Key Largo Limestone. *Chemical Geology*, 558, 119876.

PRESENTATIONS

(only showing roles as presenter or first-author abstract; ξ denotes oral talks)

13. He, R., ξ Currie, B., Kalmer, M., Ingalls, M. The effects of burial diagenesis on the stable-isotopic composition and clumped-isotope thermometry of authigenic siderite from Pennsylvanian strata in the Appalachian Basin. *GSA 2026 Annual Meeting*, Denver, CO, Oct 14th, 2026.
12. ξ He, R., Lloyd, M., Currie, B., Ingalls, M. Iron carbonate as a robust archive for reducing paleoenvironmental conditions: oxygen and clumped isotopes evidence from burial diagenesis. *The 58th Annual Graduate Student Colloquium*, Penn State Geosciences, Apr 16th, 2026.
11. He, R., Lloyd, M., Currie, B., Ingalls, M. (poster) Investigating the Role of Water-rock Alteration and Solid-state Bond Reordering on Siderite Clumped Isotopes (Δ_{47}). *Mid-Atlantic and NorthEast Geobiology Symposium*, Mantua, NJ, Feb 27th, 2026.

10. He, R., Lloyd, M., Currie, B., Ingalls, M. (poster) Investigating the Role of Water-rock Alteration and Solid-state Bond Reordering on Siderite Clumped Isotopes (Δ_{47}). *AGU 2025 Annual Meeting*, New Orleans, LA, Dec 14th, 2025.
9. He, R., Lloyd, M., Currie, B., Ingalls, M. (poster) A Systematic Assessment of Siderite Stable Isotopes ($\delta^{13}\text{C}$, $\delta^{18}\text{O}$, and Δ_{47}) Alteration During Burial Diagenesis and Late-stage Meteoric Diagenesis. *The 9th International Clumped Isotope Workshop*, Stony Brook, NY, Apr 26th, 2024.
8. $^{\epsilon}\text{He}$, R., Currie, B., Lloyd, M., Ingalls, M. An integrated assessment of siderite chemical alteration during fluid-mediated diagenesis and solid-state reordering. *AGU 2023 Annual Meeting*, San Francisco, CA, Dec 12th, 2023.
7. He, R., Currie, B., Lloyd, M., Ingalls, M. (poster) Assessment of deep burial to modern meteoric fluid-mediated diagenesis of siderite: Core-outcrop comparison from the Paleocene Fort Union Formation in Wind River Basin, Wyoming. *The 55th Annual Graduate Student Colloquium*, Penn State Geosciences, Apr 21st, 2023.
6. He, R., Currie, B., Lloyd, M., Ingalls, M. (poster) Assessment of The Impact of Late-Stage Meteoric Diagenesis on Siderite Geochemistry: Evidence from Core-Outcrop Comparison in Paleocene Fort Union Formation in Wind River Basin, Wyoming. *AGU 2022 Annual Meeting*, Chicago, IL, Dec 13th, 2022.
5. $^{\epsilon}\text{He}$, R., Lang, X.G., Zhao, K., Zhu, S.X., Li, S.Z., Shen, B. Low seawater sulfate concentration before P-T mass extinction: constrained by pyrite sulfur isotope. *The 3rd Workshop on Stable Isotope Geochemistry and Application in China*, Beijing, May 16th, 2021.
4. $^{\epsilon}\text{He}$, R., Lang, X.G., Shen, B. A rise of seawater $\delta^{13}\text{C}$ during the melting of the Marinoan glaciation. *The 19th Workshop on Algae Fossils in China*, Chengjiang, Nov 8th, 2020.
3. $^{\epsilon}\text{He}$, R., Lang, X.G., Shen, B. A rise of seawater $\delta^{13}\text{C}$ during the melting of the Marinoan glaciation. *CGU 2020 Annual Meeting*, Chongqing, Oct 18th, 2020.
2. $^{\epsilon}\text{He}$, R., Ning, M., Huang, K.J., Shen, B. Mg Isotope Systematics during Early Diagenetic Aragonite-Calcite Transition. *GSA 2018 Annual Meeting*, Indianapolis, IN, Nov 4th, 2018.
1. $^{\epsilon}\text{He}$, R., Ning, M., Huang, K.J., Shen, B. Can Mg Isotopes be used to differentiate “aragonite sea” vs. “calcite sea” in Earth’s history? *The 5th Workshop for Young Geologists in China*, Nanjing, Oct 27th, 2018.

GRANTS

Total: \$19,175

- **Collaborative Research: An isotopic and geochemical investigation of iron carbonate (siderite) as a robust archive for deep-time paleoclimate and astrobiology.**
National Science Foundation Rejected
Role: Original draft, analyses, revision, and editing
- **Ironstones in the Appalachian Basin: Understanding Siderite Clumped Isotope Solid-state Reordering.**
GSA Lipman Research Award [\$2,975] 2026
- **Ironstones in the Appalachian Basin: understanding siderite clumped isotope solid-state reordering kinetics using Paleozoic terrestrial siderite nodules.**
AAPG Grants-in-Aid Program [\$3,000] 2025

- **Ironstones in the Appalachian Basin: understanding siderite clumped isotope solid-state reordering kinetics using Paleozoic terrestrial siderite nodules.**
Sigma Xi Grants in Aid of Research [\$2,700] 2024
- **Siderite clumped isotope systematics during modern meteoric diagenesis.**
GSA Continental Scientific Drilling Division [\$2,500] 2022
- **Towards an understanding of siderite clumped isotopic behavior during burial and late-stage meteoric diagenesis.**
Mineralogical Society of America [\$5,000] 2022
- **Siderite clumped isotope systematics during late-stage meteoric diagenesis.**
Evolving Earth Foundation [\$3,000] 2022

AWARDS

Total: \$18,850

- Michael Loudin Family Graduate Scholarship [\$1,350], PSU Geosciences 2026
- Ohmoto Graduate Fellowship in Geosciences [\$1,400], PSU Geosciences 2025
- Earle S. Lenker Graduate Fellowship [\$1,400], PSU Geosciences 2025
- Parizek Graduate Fellowship Funds [\$2,400], PSU Geosciences 2024
- Katherine Freeman and Mark Patzkowsky Graduate Research Fund [\$400], PSU Geosciences 2024
- Pottorf Endowment for Graduate Excellence [\$500], PSU Geosciences 2024
- Krynine Scholarship [\$1,500], PSU Geosciences 2024
- Richard R. Parizek Graduate Fellowship [\$1,400], PSU Geosciences 2023
- Shell Geosciences Energy Research Facilitation Award [\$2,700], PSU Geosciences 2023
- Chevron Geophysics Graduate Student Scholarship [\$1,000], PSU Geosciences 2023
- Pottorf Endowment for Graduate Excellence [\$1,700], PSU Geosciences 2022
- Richard Standish Good Graduate Scholarship [\$1,600], PSU Geosciences 2022
- Charles E. Knopf, Sr. Memorial Scholarship [\$1,500], PSU Geosciences 2022

TEACHING

International Geobiology Course — Graduate Student Instructor 2023, 2024, 2025, 2026 Summer

- Instructed the CL, SEM, and EPMA modules for carbonate diagenetic screening and pyrite analyses
- Supported student final projects on Green Lake pyrite characterization, OAE II paleoenvironmental change, and travertine formation
- Analyzed organic-rich shale for source rock assessment for OAE-II samples

The Pennsylvania State University — TA 2026 Fall

- GEOSC 40 The Sea Around Us

The Pennsylvania State University — TA 2024 Fall

- GEOSC 303 Introduction to Environmental Geology
- GEOSC 105N Environments of Africa – Geology and Climate History

The Pennsylvania State University — TA 2024 Spring

- GEOSC 202 Geochemical Processes in the Geosciences
 - Taught two weekly lab sessions independently

- Designed and implemented new laboratory exercises for geochemical computing and wet-chemistry analysis
- Managed the wet chemistry lab

MENTORSHIP

Undergraduate Advising

- Mayme Kalmer (Miami University, now a master student working with Dr. Brian Currie)
- Emma Seifert (Penn State, NASA Pennsylvania Space Grant Consortium 2025 Undergraduate Research Internship Program)

PROFESSIONAL ACTIVITIES

Workshops

- International Course on Carbonate Microfacies (Flügel Course) 2024.3

Affiliations

- Sigma Xi 2026–present
- AAPG Student Member 2024–present
- Sigma Xi Affiliate Circle 2024–2026
- American Geophysical Union 2022–present
- Mineralogical Society of America 2022–present
- Geological Society of America 2018–present

PEER-REVIEW SERVICE

Palaeogeography, Palaeoclimatology, Palaeoecology (7)

FIELDWORKS

- Appalachian Basin, PA+WV+MD, USA 2024
- Lafayette Green Lake, NY, USA 2024
- Wind River Basin, WY, USA 2022
- Kuruk Tagh, Xinjiang, China 2019

SERVICE & OUTREACH

Mid-Atlantic and Northeast Geobio Symposium

Student Member of the Organizing Committee 2026

The Association for Women Geoscientists

Mentor-Mentee Program, Mentor 2024–2026

Penn State Graduate Student Committees

Perspective Students Welcoming Committee, Member 2024

TECHNICAL EXPERIENCE

- **Isotope Ratio Mass Spectrometry:** 1) Continuous flow IRMS (Thermo Scientific Delta V) coupled with Elemental Analyzer (EA) and Gas Bench (GB); 2) Dual inlet IRMS (Thermo Scientific MAT253+).
- **Inductively Coupled Plasma Optical Emission Spectroscopy (ICP-OES)** for elemental measurements.
- **IBEX Clumped Isotope System:** Automatic carbonate digestion system for carbonate clumped isotopes.

- **Source-rock analyzer (Weatherford)** for evaluating source rock maturation and thermal history. Modeling tools: Easy%Ro, Basin%Ro, pybasin.
- **Cathodoluminescence (CL), SEM, and EPMA:** Point analysis and elemental mapping for screening carbonate diagenetic alteration and mineral identification.
- **X-Ray Diffraction (XRD):** Mineralogical identification using Empyrean III system; data processing.
- **Sample preparation techniques:** 1) Carbonate digestion on glass vacuum line for carbon, oxygen, and clumped isotope analyses; 2) Organic and inorganic carbon and oxygen in rock samples; 3) DIC analyses for water samples; 4) Wet-column chromatography for carbonate and siliciclastic Mg isotope analysis.

PROGRAMMING & MODELING

R, PHREEQC, Python, MATLAB